TOPOLOGY QUALIFYING EXAM
WEDNESDAY MAY 13TH

You have three hours.

There are 7 problems, and the total number of points is 70. Show all your work. You may use standard results and theorems provided you refer to them clearly. *Please make your work as clear and easy to follow as possible.*

Name:_____

Signature:_____

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total	70	

1. (10pts) Compute the Euler characteristic of $\mathbb{RP}^n$.

2. (10pts) Let G be a finitely presented group. Show that there is a finite two dimensional CW complex whose fundamental group is isomorphic to G.

3. (10pts) Show that if G is a finitely generated group and H is a subgroup of finite index then H is finitely generated.

4. (10pts) Let X be a topological space with homology groups

$$H_k(X;\mathbb{Z}) = \begin{cases} \mathbb{Z} & \text{if } k = 0 \\ \mathbb{Z}_6 & \text{if } k = 3 \\ 0 & \text{otherwise.} \end{cases}$$

Compute the homology groups $H_\bullet(X \times \mathbb{RP}^4; \mathbb{Z})$ and $H_\bullet(X \times \mathbb{RP}^4; \mathbb{Z}_2)$

5. (10pts) Compute

$$\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Q} \oplus \mathbb{Z} \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_7, \mathbb{Z} \oplus \mathbb{Z}_6 \oplus \mathbb{Z}_{21}).$$

6. (10pts) Show that $S^2 \vee S^1 \vee S^1$ is not homotopy equivalent to $S^1 \times S^1$.

7. (10pts) Compute the cohomology ring of $\mathbb{CP}^n$.
Show the following statements are equivalent:

- n is even
- there is an orientation on $\mathbb{CP}^n$ that is preserved by every homotopy equivalence $f\colon \mathbb{CP}^n \longrightarrow \mathbb{CP}^n$.